SOFTWARE ENGINEERING LAB 6

Code and Branch Coverage Analysis

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Objective

To measure line and branch coverage for the restaurant bill application, identify untested code paths, and improve test completeness using pytest and coverage.py.

Result Summary

The initial tests passed but provided partial coverage. Additional test cases were then added. The final code coverage suite passed 20 tests and achieved 100% line coverage. The final branch coverage suite passed 22 tests and achieved 100% branch coverage.

This report includes the initial test run, baseline coverage reports, and final coverage evidence.

1. Initial Test Run

The starter test file was executed before adding additional test cases.



Figure 1. Initial starter tests for the restaurant bill application.

Observation: Both starter tests passed. More tests were needed to cover the remaining application paths.

2. Baseline Line Coverage

Coverage was measured using the original starter test file.

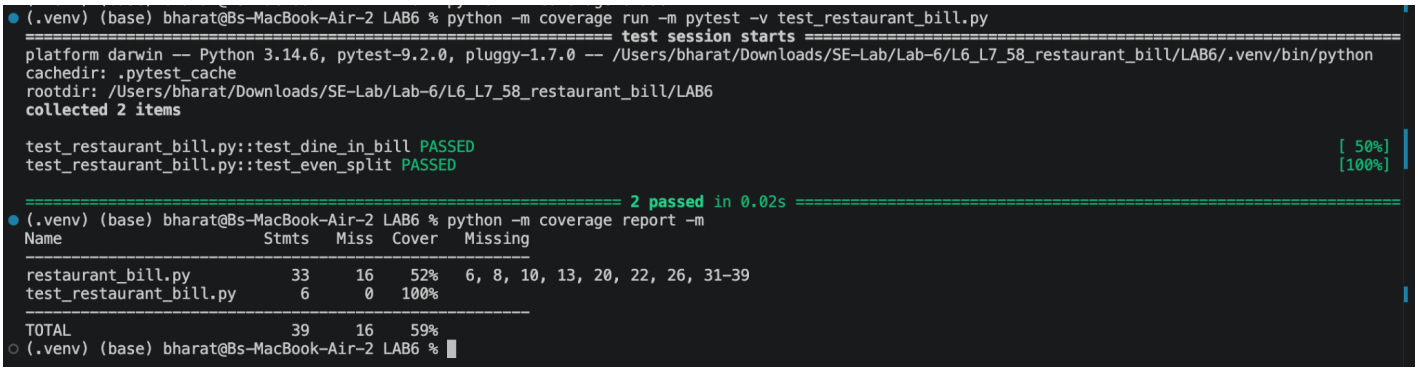


Figure 2. Initial line coverage report.

Observation: The application initially had 52% line coverage, showing that several statements were not executed.

3. Baseline Branch Coverage

Branch coverage was measured using the original starter test file and the --branch option.

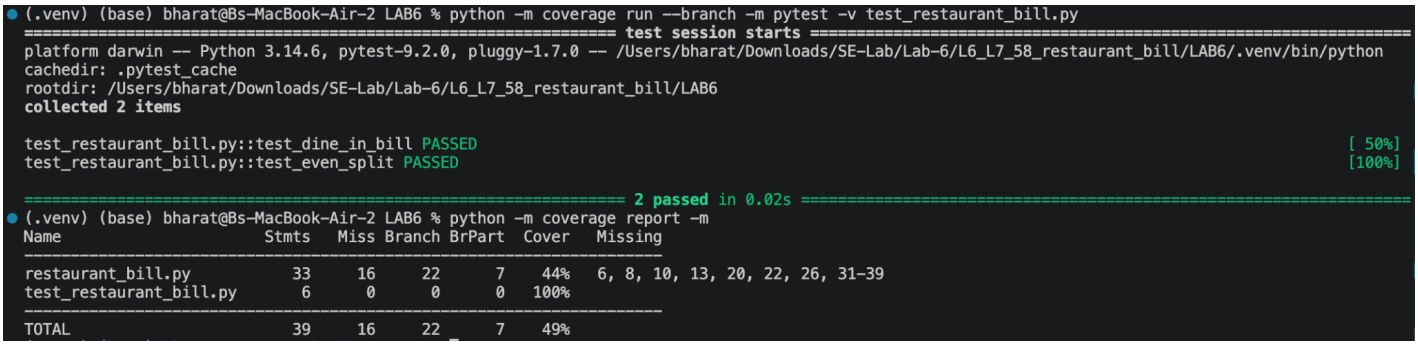


Figure 3. Initial branch coverage report.

Observation: The initial branch coverage was 44%, with multiple decision outcomes still untested.

4. Final Code Coverage

Additional test cases were added for valid inputs, invalid inputs, pricing rules, bill splitting, and seating decisions.

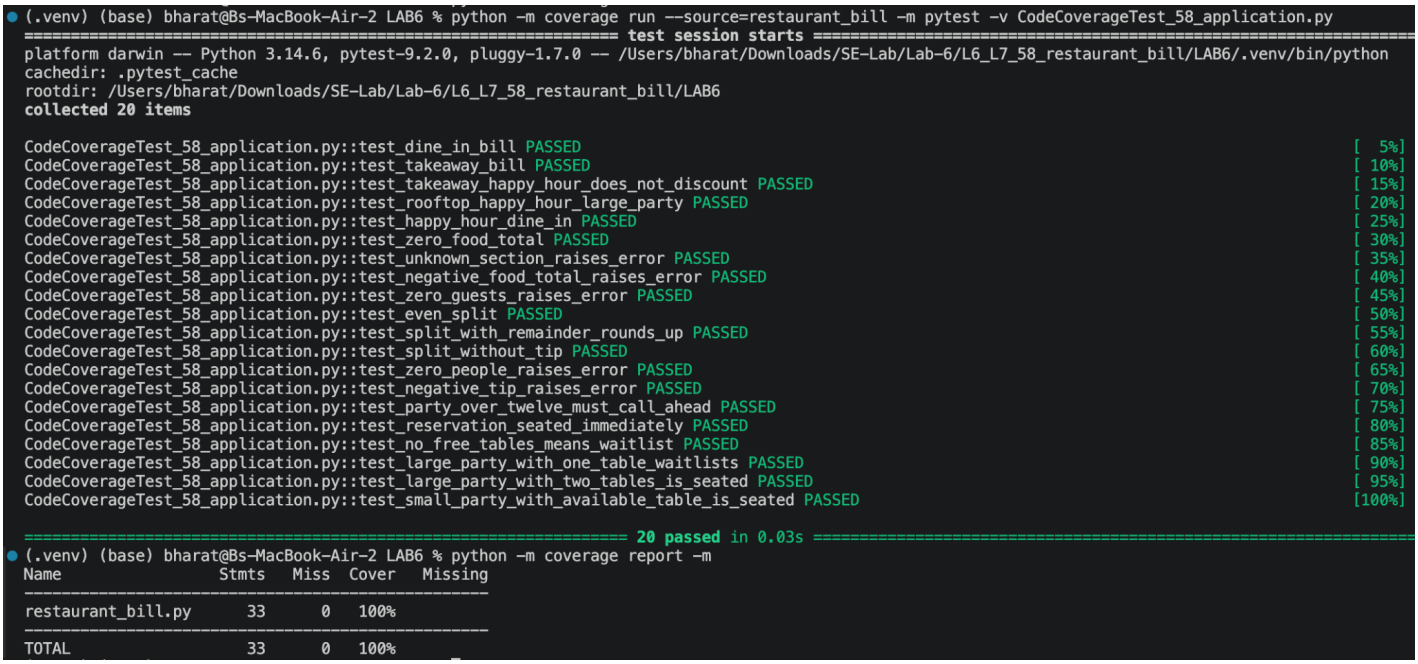


Figure 4. Final code coverage test run and report.

Observation: All 20 code coverage tests passed. The application reached 100% line coverage (33 of 33 statements).

5. Final Branch Test Execution

A separate branch coverage test suite was run to exercise both outcomes of the application's conditions.

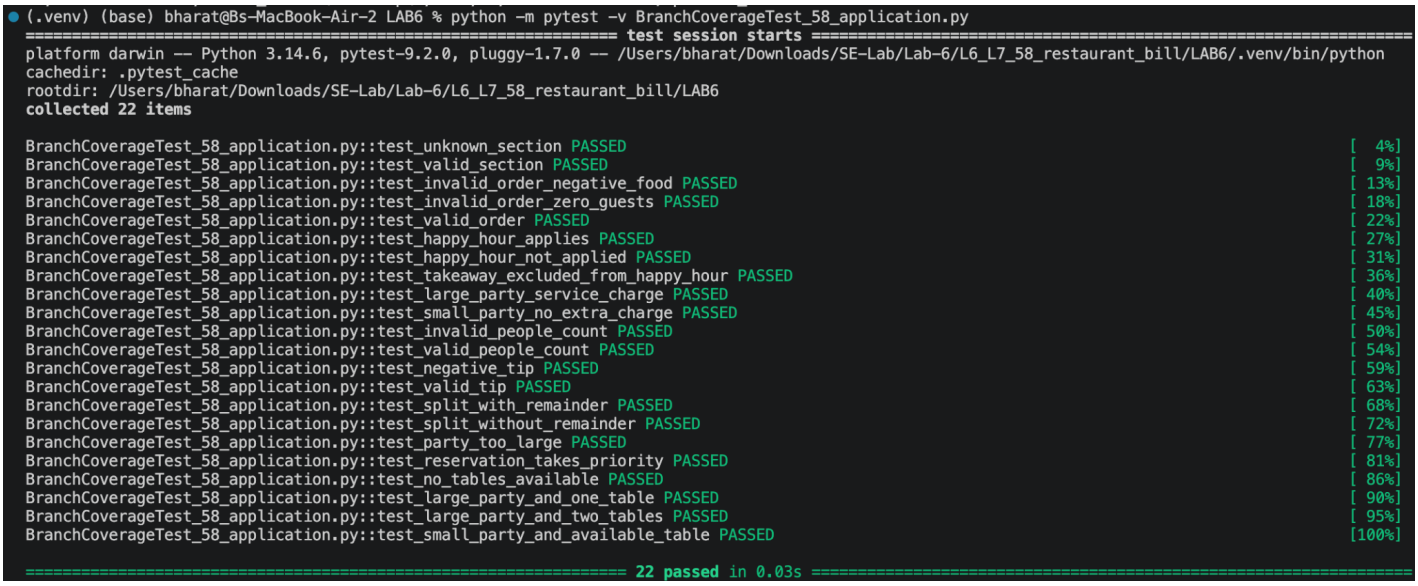


Figure 5. Branch coverage test execution.

Observation: All 22 branch coverage tests passed.

6. Final Branch Coverage Report

The final coverage report confirms the measured branch coverage for restaurant_bill.py.

```
(.venv) (base) bharat@Bs-MacBook-Air-2 LAB6 % python -m coverage run --branch --source=restaurant_bill -m pytest -v BranchCoverageTest_58_application.
py
============================================================ test session starts ============================================================
platform darwin -- Python 3.14.6, pytest-9.2.0, pluggy-1.7.0 -- /Users/bharat/Downloads/SE-Lab/Lab-6/L6_L7_58_restaurant_bill/LAB6/.venv/bin/python
cachedir: .pytest_cache
rootdir: /Users/bharat/Downloads/SE-Lab/Lab-6/L6_L7_58_restaurant_bill/LAB6
collected 22 items

BranchCoverageTest_58_application.py::test_unknown_section PASSED                                                                    [  4%]
BranchCoverageTest_58_application.py::test_valid_section PASSED                                                                      [  9%]
BranchCoverageTest_58_application.py::test_invalid_order_negative_food PASSED                                                        [ 13%]
BranchCoverageTest_58_application.py::test_invalid_order_zero_guests PASSED                                                          [ 18%]
BranchCoverageTest_58_application.py::test_valid_order PASSED                                                                        [ 22%]
BranchCoverageTest_58_application.py::test_happy_hour_applies PASSED                                                                 [ 27%]
BranchCoverageTest_58_application.py::test_happy_hour_not_applied PASSED                                                             [ 31%]
BranchCoverageTest_58_application.py::test_takeaway_excluded_from_happy_hour PASSED                                                  [ 36%]
BranchCoverageTest_58_application.py::test_large_party_service_charge PASSED                                                         [ 40%]
BranchCoverageTest_58_application.py::test_small_party_no_extra_charge PASSED                                                        [ 45%]
BranchCoverageTest_58_application.py::test_invalid_people_count PASSED                                                               [ 50%]
BranchCoverageTest_58_application.py::test_valid_people_count PASSED                                                                 [ 54%]
BranchCoverageTest_58_application.py::test_negative_tip PASSED                                                                       [ 59%]
BranchCoverageTest_58_application.py::test_valid_tip PASSED                                                                          [ 63%]
BranchCoverageTest_58_application.py::test_split_with_remainder PASSED                                                               [ 68%]
BranchCoverageTest_58_application.py::test_split_without_remainder PASSED                                                            [ 72%]
BranchCoverageTest_58_application.py::test_party_too_large PASSED                                                                    [ 77%]
BranchCoverageTest_58_application.py::test_reservation_takes_priority PASSED                                                         [ 81%]
BranchCoverageTest_58_application.py::test_no_tables_available PASSED                                                                [ 86%]
BranchCoverageTest_58_application.py::test_large_party_and_one_table PASSED                                                          [ 90%]
BranchCoverageTest_58_application.py::test_large_party_and_two_tables PASSED                                                         [ 95%]
BranchCoverageTest_58_application.py::test_small_party_and_available_table PASSED                                                    [100%]

============================================================ 22 passed in 0.03s =============================================================
(.venv) (base) bharat@Bs-MacBook-Air-2 LAB6 % python -m coverage report -m
Name                Stmts   Miss Branch BrPart  Cover   Missing
---------------------------------------------------------------
restaurant_bill.py      33      0     22      0   100%
---------------------------------------------------------------
TOTAL                   33      0     22      0   100%
```

Figure 6. Final branch coverage report.

Observation: The application achieved 100% coverage, with 22 branches covered and no missing statements or branch parts.

Conclusion

The additional tests improved coverage from 52% line coverage and 44% branch coverage to 100% for both metrics. All final tests passed successfully.